

Part 1 — Questions

****Q1.****

```
```c
```

```
int count = 0;
```

```
for(int i=0;i<4;i++)
```

```
{
```

```
 for(int j=i;j<4;j++)
```

```
 {
```

```
 count++;
```

```
 }
```

```
}
```

```
printf("%d",count);
```

```
```
```

****Q2.****

```
```c
```

```
int i = 1;
```

```
int sum = 0;
```

```
while(i <= 10)
```

```
{
```

```
 sum += i;
```

```
 i = i + 3;
```

```
}
```

```
printf("%d", sum);
```

```
``
```

**\*\*Q3.\*\***

```
``c
```

```
int a[] = {5,10,15,20,25};
```

```
int *p = a;
```

```
printf("%d ", *(p+2));
```

```
p++;
```

```
printf("%d", *(p+2));
```

```
``
```

**\*\*Q4.\*\***

```
``c
```

```
#include <stdio.h>
```

```
void fun(int n)
```

```
{
```

```
 if(n==0)
```

```
 return;
```

```
 fun(n-1);
```

```
 printf("%d ",n);
 fun(n-1);
}
```

```
fun(4);
...
```

**\*\*Q5.\*\***

```
```c  
void fun(int n)  
{  
    if(n<=0)  
        return;  
    printf("%d ",n);  
    fun(n-2);  
    printf("%d ",n);  
}
```

```
fun(7);  
...
```

****Q6.****

```
```c  
int i=1;

for(;;)
```

```
{
 if(i>20)
 break;
 printf("%d ",i);
 i=i*2+1;
}
...
```

**\*\*Q7.\*\***

```
```c  
int arr[] = {1,2,3,4};  
  
for(int i=0;i<4;i++)  
{  
    arr[i] = arr[i] * arr[3-i];  
}  
  
for(int i=0;i<4;i++)  
{  
    printf("%d ",arr[i]);  
}  
...
```

****Q8.****

```
```c  
#include <stdio.h>
```

```
void solve(){
 int x = 2;
 printf("%d", (x << 1) + (x >> 1));
}
```

```
int main() {
 solve();
 return 0;
}
...
```

**\*\*Q9.\*\***

```
```c  
int i = 0;  
  
for(i = 0; i < 10; i++)  
{  
    if(i % 2 == 0)  
        i++;  
    printf("%d ", i);  
}  
...
```

****Q10.****

```
```c
```

```
int arr[] = {2,4,6,8};
```

```
int *p = arr;
```

```
for(int i=0;i<4;i++)
```

```
{
```

```
 printf("%d ",*(p+i));
```

```
}
```

```
...
```

**\*\*Q11.\*\***

```
```c
```

```
void fun(int n)
```

```
{
```

```
    if(n<=0)
```

```
        return;
```

```
    fun(n-1);
```

```
    printf("%d ",n);
```

```
    fun(n-1);
```

```
}
```

```
fun(3);
```

```
...
```

****Q12.****

```
```c
```

```
void fun(int n)
```

```
{
 if(n==0)
 return;
 printf("%d ",n);
 fun(n-1);
 printf("%d ",n);
}
```

```
fun(3);
...
```

**\*\*Q13.\*\***

```
```c  
void fun(int n)  
{  
    if(n==0)  
        return;  
    printf("%d ",n);  
    fun(n/2);  
    printf("%d ",n);  
}
```

```
fun(8);  
...
```

****Q14.****

```
```c
```

```
int x=5;
```

```
printf("%d ",x++);
```

```
printf("%d ",++x);
```

```
```
```

****Q15.****

```
```c
```

```
int i,j=0;
```

```
for(i=1;i<=5;i++)
```

```
{
```

```
 j += i;
```

```
 if(j>6)
```

```
 break;
```

```
}
```

```
printf("%d",j);
```

```
```
```

****Q16.****

```
```c
```

```
int arr[] = {1,2,3,4,5};
```

```
int i,sum=0;
```



```
for(i=0;i<5;i++)
{
 if(arr[i]%2==1)
 sum += arr[i];
 else
 sum -= arr[i];
}
```

```
printf("%d",sum);
...
```

**\*\*Q17.\*\***

```
```c  
#include<stdio.h>  
  
int main() {  
    int i=1;  
    while(i<=10) {  
        printf("%d ",i);  
        i=i+3;  
    }  
}
```

****Q18.****

```
```c
```

```
int arr[] = {1,2,3,4,5};
```

```
int i;
```

```
for(i=0;i<5;i++)
```

```
{
```

```
 arr[i] = arr[i] + i;
```

```
}
```

```
for(i=4;i>=0;i--)
```

```
{
```

```
 printf("%d ",arr[i]);
```

```
}
```

```
...
```

**\*\*Q19.\*\***

```
```c
```

```
void fun(int n)
```

```
{
```

```
    if(n<=1)
```

```
    {
```

```
        printf("%d ",n);
```

```
        return;
```

```
    }
```

```
    fun(n-1);
```

```
    printf("%d ",n);
```

```
}
```

```
fun(4);
```

```
...
```

****Q20.****

```
```c
```

```
#include <stdio.h>
```

```
void solve(){
```

```
 char ch[10] = "abcdefghij";
```

```
 int ans = 0;
```

```
 for(int i=0;i<10;i++){
```

```
 ans += (ch[i] - 'a');
```

```
 }
```

```
 printf("%d",ans);
```

```
}
```

```
int main() {
```

```
 solve();
```

```
 return 0;
```

```
}
```

```
...
```

**\*\*Q21.\*\***

```
```c
```

```
#include<stdio.h>
```

```
int main() {  
    int i,j;  
    for(i=1;i<=3;i++) {  
        for(j=1;j<=i;j++) {  
            printf("%d ",j);  
        }  
        printf("\n");  
    }  
}
```

****Q22.****

```
``c  
int x = 5;  
int *p = &x;  
  
(*p)++;  
  
printf("%d ",x);  
  
++(*p);  
  
printf("%d",x);  
``
```

****Q23.****

```
```c
#include<stdio.h>

int main()
{
 int a=10;
 int *p=&a;
 *p=*p+5;
 printf("%d",a);
}
```
```

****Q24.****

```
```c
#include <stdio.h>
#define CUBE(x) x*x*x

void solve(){
 int ans = 216 / CUBE(3);
 printf("%d",ans);
}

int main() {
 solve();
 return 0;
}
```
```

****Q25.****

```c

#include <stdio.h>

#include <stdbool.h>

int main() {

    bool a = true;

    bool b = false;

    int x = 10;

    int y = 5;

    int p = ((x|y) + (a+b));

    printf("%d",p);

    return 0;

}

```

****Q26.****

```c

#include<stdio.h>

void fun(int n)

{

    if(n==0)

        return;

    fun(n-1);

    printf("%d ",n);

```
}
```

```
int main()
```

```
{
```

```
 fun(4);
```

```
}
```

```
...
```

**\*\*Q27.\*\***

```
```c
```

```
#include <stdio.h>
```

```
void solve(){
```

```
    int a[] = {1,2,3,4,5};
```

```
    int sum = 0;
```

```
    for(int i=0;i<5;i++){
```

```
        if(i%2==0){
```

```
            sum += *(a+i);
```

```
        } else{
```

```
            sum -= *(a+i);
```

```
        }
```

```
        printf("%d\n",sum);
```

```
    }
```

```
}
```

```
int main() {
```

```
    solve();
```

```
    return 0;
}
...
```

****Q28.****

```
```c
void fun(int n)
{
 if(n==0)
 return;
 printf("%d ",n);
 fun(n-1);
 printf("%d ",n*n);
}
```

```
fun(3);
...
```

**\*\*Q29.\*\***

```
```c
int i,j;

for(i=1;i<=3;i++)
{
    for(j=i;j<=3;j++)
    {
```



```

        printf("%d ",j);
    }
}
...

**B1.**

```c
#include <stdio.h>

int main() {
 int x = 10;
 if(x==10){
 printf("x is 10\n");
 } else{
 printf("x is not 10\n");
 }
 return 0;
}
...

```

```

B2.

```c
#include <stdio.h>

void changeValue(int *a) {
    *a = 100;
}

```

```
int main() {  
    int x = 10;  
    changeValue(&x);  
    printf("x = %d\n", x);  
    return 0;  
}  
...
```

****B3.****

```
```c
```

```
#include <stdio.h>
```

```
void swap(int *a, int *b) {
 int temp = *a;
 *a = *b;
 *b = temp;
}
```

```
int main() {
 int x = 10, y = 20;
 swap(&x, &y);
 printf("x = %d, y = %d\n", x, y);
 return 0;
}
...
```

**\*\*B4.\*\***

```c

#include <stdio.h>

int main() {

int a = 10, b = 20, c;

c = a + b;

printf("Sum = %d\n", c);

return 0;

}

```

**\*\*B5.\*\***

```c

#include <stdio.h>

void func(int *p, int *q){

p = q;

*p = 2;

}

int main() {

int a = 1, b = 3;

int *p = &a, *q = &b;

func(p, q);

printf("%d %d\n", a, b);

```
    return 0;
}
...

```

****B6.****

```
```c
#include <stdio.h>

int main() {
 int x = 10;
 if (x == 10) {
 printf("x is 10\n");
 } else {
 printf("x is not 10\n");
 return 0;
 }
}
...

```

**\*\*B7.\*\***

```
```c
#include <stdio.h>

int main() {
    int x = 10;
    printf("x = %d\n", x);
}

```

```
    return 0;
}
...
```

****B8.****

```
```c
```

```
#include <stdio.h>
```

```
int main() {
 int x = 5;
 if(x>1){
 printf("x is greater than 1\n");
 if(x < 10){
 printf("x is less than 10\n");
 }
 } else{
 printf("x is 1 or less\n");
 }
 return 0;
}
...
```

**\*\*B9.\*\***

```
```c
```

```
#include <stdio.h>
```

```
void func(int *p) {
```

```
    *p = 20;
```

```
}
```

```
int main() {
```

```
    int a = 10;
```

```
    func(&a);
```

```
    printf("a = %d\n", a);
```

```
    return 0;
```

```
}
```

```
...
```

****B10.****

```
```c
```

```
#include <stdio.h>
```

```
int main() {
```

```
 int x = 5;
```

```
 printf("%d %d %d\n", x, x++, x);
```

```
 return 0;
```

```
}
```

```
...
```

**\*\*B11.\*\***

```
```c
```

```
#include <stdio.h>
```

```
int main() {  
    int x = 10, y = 20, z = 30;  
    printf("%d %d %d\n", x, y, z);  
    return 0;  
}  
...
```